Architecture Selection and Justification

Second-Hand Product Escrow Marketplace

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Architectural Choice. We chose Microservices with an API Gateway for the Second-Hand Product Escrow Marketplace. The system is built from services that deploy on their own: Order Manager, Payment Service, Escrow Ledger, Inspection Service, Notification Service and Identity and Access. One Web API Gateway faces the internet. The buyer app and the inspection agent console talk only to that gateway.

Reason One: Each Part Scales on Its Own. Work here is uneven. Browsing listings, checking order status and uploading inspection photos grow with the number of users, while money operations are few but must never be wrong. Requirement NFR-002 asks for the escrow lock and status update to finish in under 2 seconds for 500 concurrent users. With separate services we add copies of the gateway and the Inspection Service when the crowd grows, and keep the Payment Service and Escrow Ledger small. In one deployable unit, the same growth would force copies of the whole application, money code included.

Reason Two: A Slow Part Cannot Stop the Money Path. Requirement NFR-001 says a dispute must freeze escrow funds at once and block withdrawals while the case is open. That freeze must not queue behind an email backlog or a slow photo upload. Because Notification Service and Inspection Service run apart from the money path, neither can delay it, and the Order Manager reaches the Escrow Ledger directly, so the freeze still works when notifications are down. In a client-server design the same incident would hold the whole system.

Security Advantage. The Payment Service and the Escrow Ledger own their data, and no other service can write to the ledger. Card details never leave the Payment Service, which is the only component that holds the payment gateway credential, so a weak spot in the inspection or notification code cannot expose card data or move money. The gateway is the only internet facing component, and calls between services are mutually authenticated. The ledger is append only, so every hold, freeze, release and refund is traceable and cannot be edited quietly, which is what the dual verification rule of FR-001 needs.

Performance Benefit. Notifications, reminders and the 48 hour auto rule travel as asynchronous events, outside the purchase path, so the buyer never waits for an email. The synchronous chain for one purchase is short: gateway, Order Manager, Payment Service, Escrow Ledger. Only these four services sit inside the 2 second target of NFR-002, and each can be scaled or tuned on its own.

Tradeoff We Accept. Microservices cost more to run: more deployments to watch, more network hops, and event delivery between services that is eventually consistent. We accept this because the escrow rules here are strict, and we lower the cost by keeping the money path short, sending every client call through one gateway, and tagging each request with a single trace id.